Optimizing Corporate Banking with the Rocket Account Solution

Executive Summary

This case study explores how a leading financial institution used a custom-built innovative corporate banking solution, known as the Rocket Account. The project aimed to address the challenge of optimizing corporate financial management, specifically focusing on enhancing liquidity and maximizing earnings. By implementing a tailored solution that integrated advanced features such as automated fund sweeps into fixed deposits and comprehensive financial insights, the institution successfully improved client engagement, operational efficiency, and overall financial performance.

Client’s need for the solution

The client required a web-based solution to manage corporate accounts effectively, allowing businesses to optimize their idle funds without sacrificing liquidity. The need for a more automated and efficient financial management system led to the development of the Rocket Account.

What challenge did they face?

The primary challenge was managing idle funds in corporate accounts, which often sat unused and did not generate returns. The client needed a solution that could automatically sweep these funds into fixed deposits, ensuring that every rupee worked towards generating income while still being accessible for transactional needs.

The Rocket Account solution effectively addressed the challenge by introducing automated fund management features. It allowed corporate clients to sweep surplus balances into fixed deposits automatically, enhanced liquidity management, and provided detailed financial insights, thereby solving the client's challenge.

About the Solution

The decision-making process involved a thorough understanding of the client’s requirements for a seamless financial management system. Key features, such as automated fund sweeps and detailed financial reporting, were prioritized to enhance the client's product offering. The strategy was to develop a robust, scalable architecture that could support the complex needs of corporate clients.

Tech Stack Used

- Java
- Next JS
- Microservices Architecture
- PostgreSQL
- JWT
- JUnit 5

The project began with extensive requirement gathering, which involved coordination with multiple teams, including the Central Banking System (CBS), Optimus, and Mule teams. The solution was designed to be modular, with key components like Cel-Auth for authentication, Cel-Orch for intermediate processing, and Cel-Schedule for managing database operations and cron jobs.

Key Modules of the Solution:

- **Cel-Auth** : Verifies user authentication by validating tokens and assigning additional roles on top of those provided by the BXP service, ensuring secure access to sensitive financial data.
- **Cel-Orch** : Orchestrates interactions by consuming data from various sources and returning it to users, facilitating smooth communication between different services and modules.
- **Cel-Schedule** : Manages scheduled jobs and processes Kafka-related data, handling key tasks such as fund sweeps and report generation.

Widget Details of the Rocket Account:

The Rocket Account widget provides corporate account holders with essential financial details:

- Rocket account numbe
- Rocket account balance
- Current account (CA) balance
- Fixed deposit (FD) balance
- Incremental earnings
- Custom earnings
- Sweep-in threshold
- Projected earnings

These details are fetched from the Cel-Schedule and Optimus services. Additionally, the deep-dive arrow allows users to explore:

- Weekly and monthly transaction charts
- Rocket returns (YTD, interest accrued, interest earned, hold funds, threshold value)
- Active and closed FD details (FD number, creation/maturity date, TDS, FD status)

Borrowing Summary:

Corporate account holders can also view borrowing details via dedicated widgets:

- Term Loan (TL)
- Short Term Loan (STL)
- Overdraft (OD)
- Cash Credit (CC)
- Working Capital Demand Loan (WCDL)

API and Third-Party Integration:

- **API Integration** : All API requests include a bearer token, which extracts user details (ID, permissions, roles, accessible accounts).
- **Third-Party API Integration** : The Optimus Rest Template is used for calling internal services.

Testing:

- **JUnit Test Cases** : JUnit test cases are used to verify the Java code’s functionality, maintaining a 95% test case ratio for each service and controller.

Types of Users:

- **Corporate Account Holders** : The primary users who benefit from automated financial management and detailed reporting.
- **Bank Administrators** : Users who manage and oversee the operation of the Rocket Account system.

Benefits of the Solution:

Automated Fund Management: Surplus balances are automatically swept into fixed deposits, maximizing earnings.

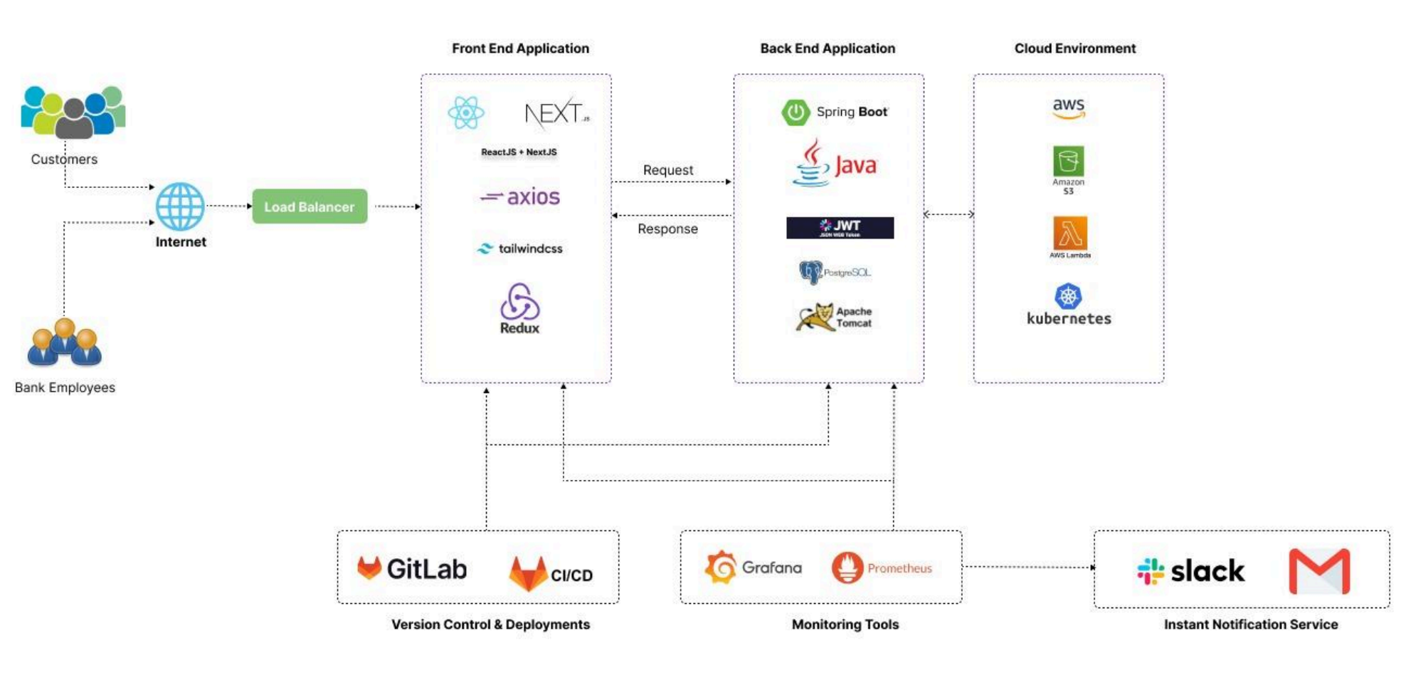
Enhanced Financial Insights : Detailed reporting on weekly and monthly transactions, returns, and fixed deposit statuses.

Improved Client Engagement : The Rocket Account offers a user-friendly interface with comprehensive financial management features.

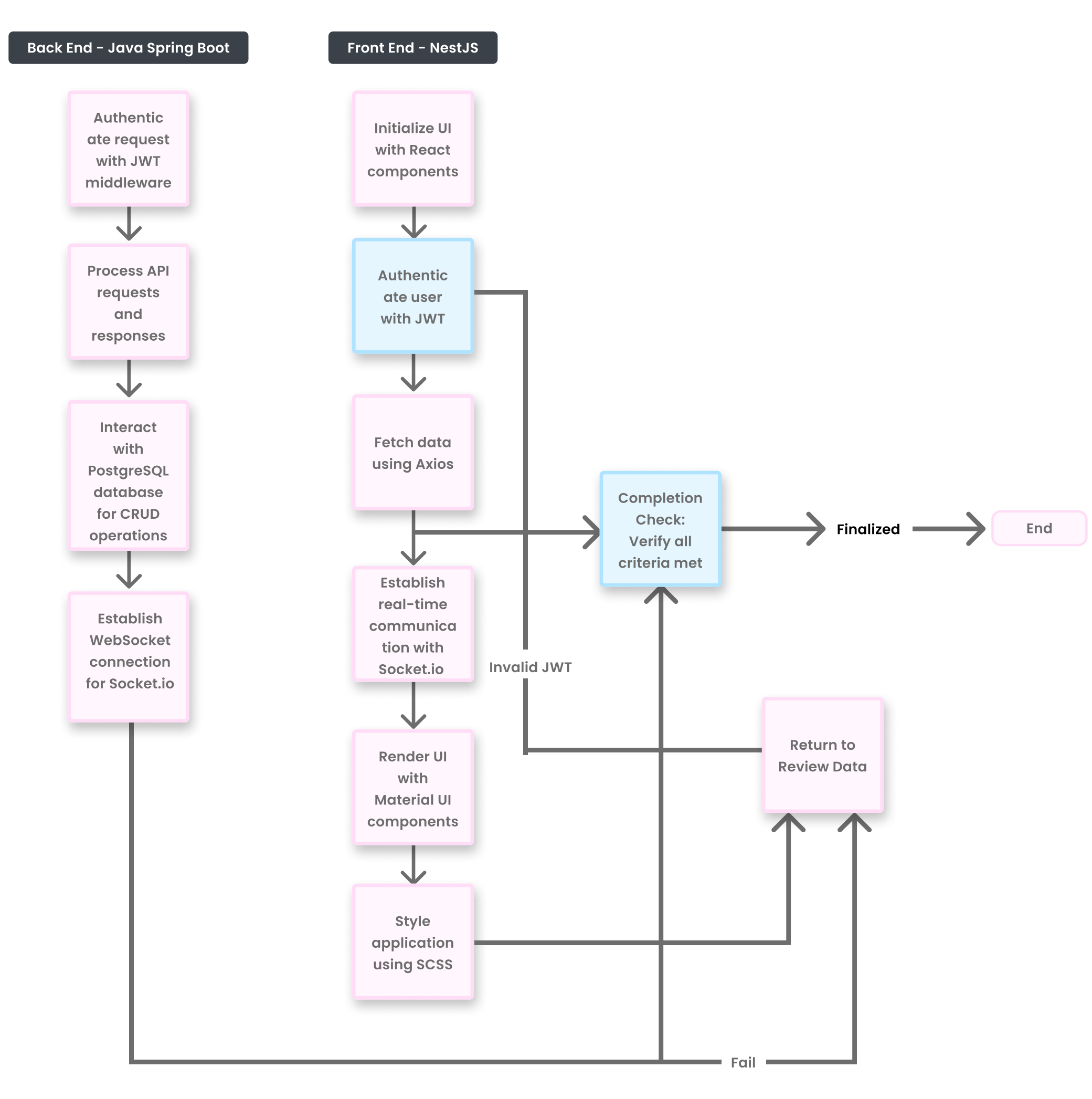
Soft Suave’s Challenges

The main challenge was gathering the necessary requirements due to the involvement of multiple teams, including CBS, Optimus, and Mule. This coordination was crucial to ensure the solution met the client’s needs.
Despite the complexity, the challenges were overcome through persistent communication and collaboration with the various teams involved. The development team maintained a flexible approach, adapting to new information and requirements as they emerged.

Architecture Design



Flow Diagram


Back End - Java Spring Boot
Authenticate request with JWT middleware
Process API requests and responses
Interact with PostgreSQL database for CRUD operations
Establish WebSocket connection for Socket.io
Front End - NestJS
Initialize UI with React components
Authenticate user with JWT
Fetch data using Axios
Establish real-time communication with Socket.io
Render UI with Material UI components
Style application using SCSS
Completion Check: Verify all criteria met
Finalized
End
Invalid JWT
Return to Review Data
Fail

About Soft Suave

Soft Suave is a prominent IT outsourcing and software development company with a rich history of serving global clients for more than eleven years. with our core expertise in IT staff augmentation, we assist businesses in augmenting their existing workforce by providing skilled professionals. our team consists of talented, knowledgeable, and dynamic developers who are devoted to offering revolutionary and effective technology solutions. we offer tailored solutions that meet the unique needs of our clients, and our customer-centric approach ensures that our clients are involved at every stage of the project.

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